



TXW80X audio solution development guide



Notice

Due to product version upgrades or other reasons, this document will be updated from time to time. Unless otherwise agreed, this document only

As a guide for use without any warranty.

Zhuhai Taixin
Semiconductor Co.,Ltd

3rd Floor, Building 11, Science and Technology Innovation Park, No. 1, Gangwan, High-tech Zone, Zhuhai City

Confidentiality level	A	TXW80X SDK Quick Start Manual	File No	TX-0000
Release date 2022-08-08			file version	V1.0

Revision history

date	Version	describe	Revised by
2022-08-08	V1.0	First version, note: only versions after SDK V18676 are supported Support Bluetooth distribution network function	TX



Zhuhai Taixin Semiconductor Co., Ltd.
Zhuhai Taixin Semiconductor Co.,Ltd

3rd Floor, Building 11, Science and Technology Innovation Park, No. 1, Gangwan, High-tech Zone, Zhuhai City

Copyright infringement will be investigated

Copyright © 2021 by Tai Xin All rights reserved

Confidentiality level	A	TXW80X SDK Quick Start Manual	File No	TX-0000
Release date 2022-08-08			file version	V1.0
Table of contents 1 Overview..... 1 2. Hardware development board..... 1 2.1. Audio and video development board..... 2 2.1.1. Audio and video development board interface introduction2 3. Bluetooth network configuration process..... 3 3.1. Bluetooth distribution network related configuration.....3 3.2. Bluetooth distribution network demo..... 3 3.3. Small wireless transmission..... 5				
		Zhuhai Taixin Semiconductor Co., Ltd. Zhuhai Taixin Semiconductor Co.,Ltd		3rd Floor, Building 11, Science and Technology Innovation Park, No. 1, Gangwan, High-tech Zone, Zhuhai City
Copyright infringement will be investigated Copyright © 2021 by Tai Xin All rights reserved				

1 Overview

This article mainly describes the Bluetooth distribution network process.

This document is mainly applicable to the following engineers:

- Technical support engineer
- Solution software development engineer

Product scope applicable to this document:

model	encapsulation	Package
TXW806	QFN56/48/40	

2. Hardware development board

For quick start and solution evaluation, we provide development boards for various application scenarios.

2.1. Audio and video development board

2.1.1. Audio and video development board interface introduction

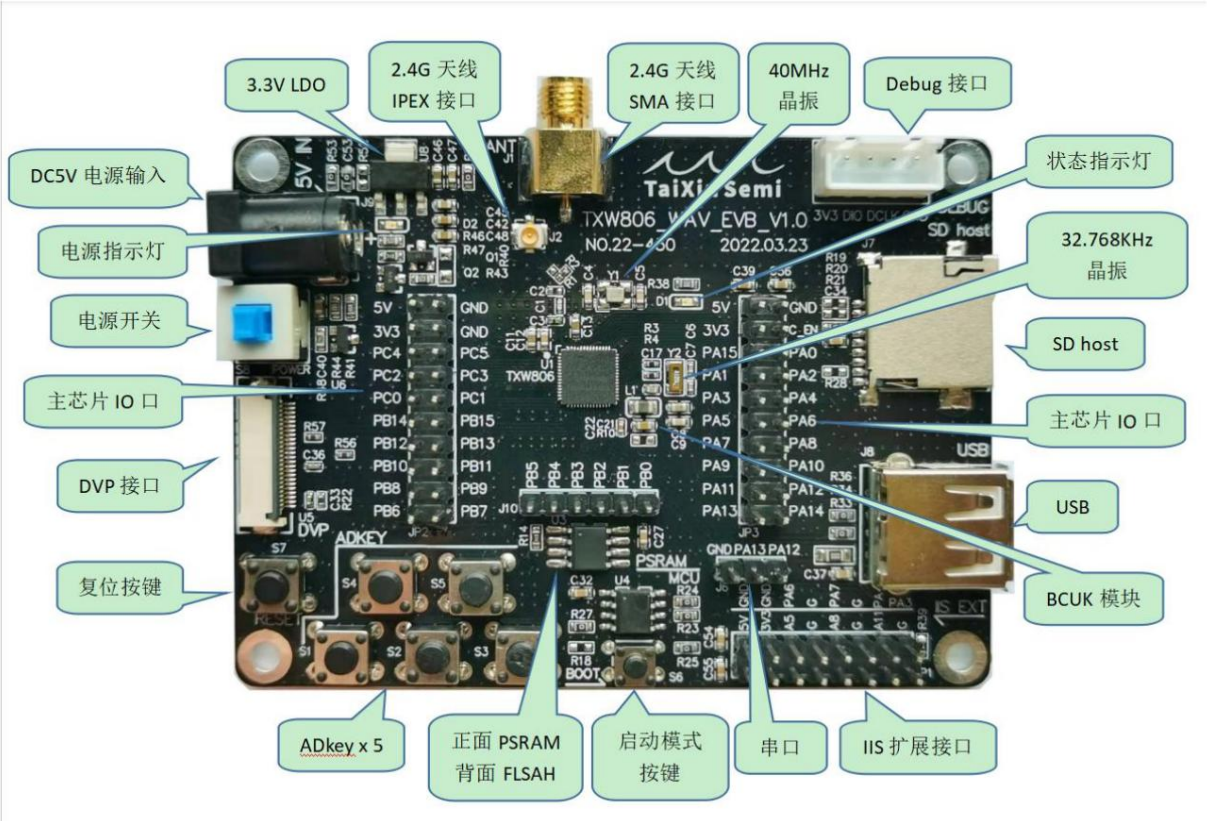


Figure 3.1.1.1 - Main view of audio and video development board

Special Instructions

Mode start button: This button can save the system with one click, run to death when the chip is powered on, cklinc programming and other upgrades

Used when all levels are disabled.

IIS extension interface: This interface is used to expand the audio daughter board.

3. Bluetooth configuration process

Note: Before starting the Bluetooth distribution network, the wifi related configuration must be initialized.

3.1. Bluetooth distribution network related configurations

The project configuration mainly depends on the project_config.h configuration.

```
3  #define BLE_PAIR_NET 1
```

Set the macro BLE_PAIR_NET to 1 and open the demo of the distribution network by default (adaptive BLE distribution network applet) Bluetooth distribution network

demo

```
#ifdef CONFIG_UMAC4
    #if BLE_PAIR_NET == 1
        //配网仅支持station模式
        if(sys_cfgs.wifi_mode == WIFI_MODE_STA)
        {
            //启动蓝牙配网
            //没有配过网络,先进行网络配置
            if(!get_sys_cfgs_ble_pair_status())
            {
                ble_reset_pair_network(ble_demo_recv_data);
            }
            //否则直接发送小无线
        }
        else
        {
            //oudi, 2周前 * 1、重新修改ble配网,并且添加小无线发送,参考ble_pair...
            ble_reset_pair_network(NULL);
        }
    }
#endif
#endif
```

```

1  * *****/
2  int8 ble_demo_rcv_data(void *priv,uint8 *data,int len)
3  {
4      #if 0
5          for(int i=0;i<len;i++)
6          {
7              if(i%16 == 0)
8              {
9                  printf("\r\n");
10             }
11             printf("%02X ",data[i]);
12         }
13         printf("\r\n");
14         //return RET_OK;
15     #endif
16
17     struct ble_data *ssid_pass = (struct ble_data*)data;
18     if(ssid_pass->type == 1)
19     {
20         memcpy(ble_ssid,data+sizeof(struct ble_data),ssid_pass->size);
21         data += (ssid_pass->size+sizeof(struct ble_data));
22     }
23     else
24     {
25         return RET_ERR;
26     }
27
28     ssid_pass = (struct ble_data*)data;
29     if(ssid_pass->type == 2)
30     {
31         memcpy(ble_password,data+sizeof(struct ble_data),ssid_pass->size);
32         //识别完后,type=3没有去判断,自动wifi去适配 ends, 2周前 1、修改阿里云的demo(原版触摸控制led)这些可能就不合...
33         ble_wifi_station_reset(ble_ssid,ble_password);
34         return RET_OK;
35     }
36     else
37     {
38         return RET_ERR;
39     }
40
41     return RET_ERR;
42 }

```

1. get_sys_cfgs_ble_pair_status Gets whether the Bluetooth network interface has been configured (if it has been configured through Xiaocheng

The configuration network information is set by the program and will be saved for the next boot. If you need to re-distribute the network, you need to modify the status.

Change, the demo does not have the network reconfiguration operation)

2. If you need to customize the distribution network protocol, you need to re-implement the ble_demo_rcv_data interface yourself.

Just identify the ssid and password according to the protocol, and register through ble_reset_pair_network

3. int8 ble_demo_rcv_data(uint8 *data,int len), the custom protocol needs to be re-implemented.

port, data is Bluetooth broadcast packet data (debugging can print the contents of data), len is the length, and the return values are

RET_OK and RET_ERR, if ssid and password are recognized, RET_OK will be returned, otherwise RET_OK will be returned.

RET_ERR, when returning RET_OK, it will automatically exit the Bluetooth distribution mode.

4. ble_wifi_station_reset(ble_ssid,ble_password), the Bluetooth distribution network interface, if recognized

ssid and password, call this interface and return RET_OK, so that the corresponding station will be connected.

And save the corresponding parameters. Next time you turn on the computer, you will directly connect to the corresponding station. If the network configuration fails, you need to

To re-call the thread for network distribution (screenshot part of the code)

3.3. Small wireless transmission

```
void ble_reset_pair_network(void *cb)
{
    //发送小无线信号
    if(!cb)
    {
        start_ble_network(g_ops, NULL, 38, NULL);
        lmac_ble_tx(g_ops, tx_payload_gfsk, sizeof(tx_payload_gfsk));
        stop_ble_network(g_ops);
    }

    //进入蓝牙配网
    else
    {
        int err = start_ble_network(g_ops, NULL, 38, (recv)cb);
        if(err == RET_OK)
        {
        }
    }
}
```

As shown in the red box:

1. The small wireless transmission is sent through the Bluetooth channel. You can use the small wireless to receive the corresponding data. The data formats are all

It is customized. The data sent is tx_payload_gfsk. Fill it in according to the actual content.